

WEEK 1 AI ASSIGNMENT REPORT

Q1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a technology that allows computers and machines to perform tasks that normally require human intelligence. AI can learn from data, recognize patterns, understand language, solve problems, and make decisions.

Today, AI is used in many applications such as chatbots, voice assistants, recommendation systems, self-driving cars, and image recognition. AI helps people complete tasks faster and more efficiently.

Examples of AI:

- ChatGPT
- Google Assistant
- Siri
- Netflix recommendations

AI is becoming an important part of our daily lives and is helping businesses, students, and professionals in many ways.

Q2. Difference Between AI, Machine Learning, Generative AI, and Agentic AI

Artificial Intelligence (AI) is the broad field of creating smart machines that can perform intelligent tasks.

Machine Learning (ML) is a part of AI where computers learn from data and improve their performance without being explicitly programmed.

Generative AI is a type of AI that can create new content such as text, images, music, videos, and code. ChatGPT and image generators are examples of Generative AI.

Agentic AI is a more advanced form of AI that can plan tasks, make decisions, and take actions to achieve a goal with minimal human guidance.

Simple Comparison:

AI → Makes machines intelligent.

Machine Learning → Learns from data.

Generative AI → Creates new content.

Agentic AI → Acts independently to complete tasks.

Q3. Explain Any Three Open-Source AI Models Available on Hugging Face.

1. Llama

Llama is an open-source language model developed by Meta. It is used for chatbots, content generation, coding assistance, and research. Many developers use Llama because it is powerful and freely available.

2. Mistral

Mistral is a fast and efficient open-source AI model. It can answer questions, summarize text, and generate content. It is known for providing good performance while using fewer resources.

3. FLAN-T5

FLAN-T5 is an open-source model developed by Google. It can perform tasks such as translation, summarization, question answering, and text generation. It is widely used in Natural Language Processing (NLP) projects.

Q4. Difference Between Traditional Programming and AI-Assisted Development

Traditional programming requires developers to write all instructions and logic manually. The computer follows the exact rules written by the programmer.

AI-assisted development uses AI tools to help developers write code, suggest solutions, find bugs, and improve productivity.

Traditional Programming:

- Developer writes all code manually.
- Fixed rules are followed.
- Takes more time.

AI-Assisted Development:

- AI helps generate code.
- Faster development.
- Improves productivity.
- Helps find errors and suggestions.

AI-assisted development allows programmers to focus on solving problems while AI handles repetitive tasks.

Q5. Explain AI Hallucinations with One Example

AI hallucination happens when an AI system gives incorrect or made-up information while sounding confident.

This happens because AI predicts answers based on patterns in data and may sometimes generate information that is not true.

Example: If a user asks an AI chatbot about a book written by a famous author, the AI might create a fake book title that does not actually exist. The answer may look correct, but it is inaccurate. Therefore,

users should always verify important information provided by AI systems.

